LIU HENGZHI

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1. EDUCATION

National University of Singapore **2026 – Present**
Bachelor of Computing in Computer Science *Singapore*

Temasek Polytechnic **2023 – 2026**
Diploma in Biomedical Engineering with Merit (GPA: 3.91/4.00) *Singapore*

- Polytechnic Scholarship (Engineering); Course Bronze Medalist; Director's List
- Focused on embedded systems, hardware-software co-design, and applied engineering research

2. SKILLS

Programming Languages: Python, C++, C, Swift

Systems & Backend: Embedded Systems, Real-time Systems (FreeRTOS), System Design Basics, Firmware Development

Hardware & Edge Systems: ESP32, Arduino, PCB Design (LCEDA), Sensor Integration, Edge Computing Concepts

Tools: Git/GitHub, Linux (basic), CAD (Fusion 360 / NX / Creo), Adobe Suite

3. WORK EXPERIENCE

National University of Singapore (iHealthtech) **Mar 2025 – Jul 2025**
Research Engineering Intern (Embedded & Biomedical Systems) *Singapore*

- Contributed to development and testing of embedded biomedical prototype systems for experimental data acquisition and validation
- Worked on hardware instrumentation workflows, improving stability and repeatability of experimental setups
- Assisted in debugging and calibration of sensor-driven systems used in biomedical research environments
- Documented experimental pipelines and ensured reproducibility of hardware test procedures

4. TECHNICAL PROJECTS

Smart Mahjong Play-Board (Embedded Systems / Edge Computing Prototype) **2026**

Independent Systems Developer *C++, ESP32, PCB Design, Embedded Firmware*

- Designed an embedded edge-computing system using ESP32 for real-time event detection and state tracking
- Implemented firmware in C++ for low-latency input processing and hardware-software integration
- Designed PCB layout (LCEDA) optimizing power distribution and reducing physical footprint constraints
- Developed system-level architecture integrating sensors, microcontroller logic, and external I/O modules

Experiential Research Programme (Fuel Cell Materials Research) **2024**

Research Student

- Conducted experimental investigation on titanium electroplating for corrosion resistance in fuel cell components
- Performed materials testing and analysis to evaluate structural stability under chemical exposure conditions
- Worked with lab instrumentation and iterative experimental design to improve coating consistency

5. TEACHING EXPERIENCE

Chinese Teacher (Part-time) **Apr 2024 – Dec 2024**

- Developed structured communication and mentoring skills through one-on-one and group instruction

6. UNIVERSITY ENGAGEMENT

- Incoming Undergraduate Student at School of Computing, National University of Singapore.
- Active engagement in computing community frameworks and prospective engineering alignment programs.

7. CO-CURRICULAR ACTIVITIES

WorldSkills Singapore (Additive Manufacturing) **2024**

- 1st Place (Qualifying Round); demonstrated advanced CAD modeling, additive manufacturing optimization, and engineering precision design

Electronic Design & Robotics Competitions **2024**

- Developed embedded logic systems and real-time control algorithms for robotics and circuit design challenges

8. ADDITIONAL INFORMATION

- **Leadership:** Temasek LEAD & ENGenius Excellence Programs alumnus
- **Languages:** English (Professional), Mandarin (Native)
- **Interests:** Distributed systems, AI infrastructure, embedded systems, scientific computing